Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[ ]beamerthemeNamkeen
[utf8]inputenc [english]babel [T1]fontenc helvet

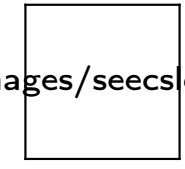
The Namkeen Beamer Theme

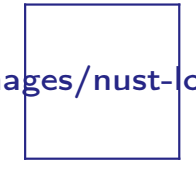
v. 1.0

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19 Pebruari 2025





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Introduction

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Introduction

Why the Namkeen beamer theme?

- In October 2020, I had to give a presentation at an International Conference.
- Since there was no NUST School of Electrical Engineering and Computer Science (NUST-SEECS) branded beamer theme, I tried to create the Namkeen beamer theme.
- This theme is based on the open source AAU sidebar theme so that other researchers and students at NUST could use the theme for their presentations.



Introduction

License

- I have used the logo from . As long as you use the theme for making presentations in connection with your work at NUST, you are allowed to use the NUST logo.



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License

- I have used the logo from . As long as you use the theme for making presentations in connection with your work at NUST, you are allowed to use the NUST logo.
- The rest of the theme is provided under the GNU General Public License v. 3 (GPLv3). This basically means that you can redistribute it and/or modify it under the same license. For more information on the GPL license see

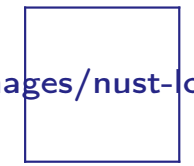


Installation

The theme consists of four files

1. `beamerthemeNamkeen.sty`
2. `beamerinnerthemeNamkeen.sty`
3. `beamerouterthemeNamkeen.sty`
4. `beamercolorthemeNamkeen.sty`

The theme can either be installed for local or global use.



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The theme can either be installed for local or global use.

Local Installation

The simplest way of installing the theme is by placing the four theme files in the same folder as your presentation. When you download the theme, the four theme files are located in the `local` folder.



Installation

Global Installation

- If you wish to make the theme globally available, you must put the files in your local latex directory tree. The location of the root of the local directory tree depends on the operating system and the latex distribution. On the following slides, you can read the instructions for some common setups.
- When you download the theme, the four theme files are embedded in a directory structure (in the `global` folder) ready to be copied directly to the root of your local directory tree.
- On the following slides, we refer to this directory structure as `<dirstruct>`. Note that some parts of the directory may already exist if you have installed other packages in your local latex directory tree. If this is the case, you simply merge `<dirstruct>` with your existing setup.

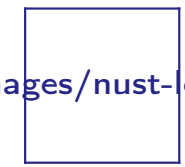


Installation

GNU/Linux

Ubuntu with TeX Live

1. Place the `<dirstruct>` in the root of your local latex directory tree. By default it is
   `~/texmf`
   If the root does not exist, create it. The symbol ~ refers to your home folder, i.e., `/home/<username>`
2. In a terminal run
   `$ texhash ~/texmf`



Windows with MiKTeX

Apparently, MiKTeX does not include a local latex directory tree by default. Therefore, you first have to create it.

1. To do this, create a folder `<somewhere>` named, e.g., `texmf`
2. Add this folder in the Roots tab of the MiKTeX Settings dialog
3. Place the `<dirstruct>` in your newly created local latex directory tree
   `<somewhere>\texmf`
4. Open the MiKTeX Settings dialog and click Refresh FNDB.



Windows with TeX Live

In the advanced TeX Live Installer, you can manually change the default position of the root of the local latex directory tree. However, we assume the default position below.

1. Place the `<dirstruct>` in your local latex directory tree `%USERPROFILE%\texmf`
   If it does not exist, create it. In XP `%USERPROFILE%` is `c:\Document and Settings\<username>`
   by default, and in Vista and above it is by default `c:\Users\<username>`
2. Open the TeX Live Manager dialog and select 'Update filename database' under 'Actions'.

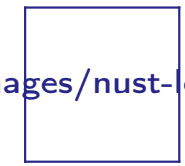


Installation

Mac OS X

Mac OS X with MacTeX

Place the `<dirstruct>` in the root of your local latex directory tree. By default it is
`~/Library/texmf`
If the root does not exist, create it. The symbol ~ refers to your home folder, i.e., `/home/<username>`



Installation

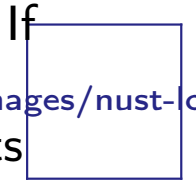
Required Packages

Of course, you have to have the Beamer class installed. In addition, the theme loads two packages

- TikZ[1]
- calc

These packages are very common and should therefore be included in your latex distribution.

[1] By the way, TikZ is an awesome package for creating beautiful graphics. If you do not believe me, then have a look at these or the . If you want to create beautiful plots, you should use the pgfplots package which is based on TikZ.



The Presentation Theme

It is very simple to load the presentation theme. Just type
`\usetheme[<options>]{Namkeen}`
which is exactly the same way other beamer presentation themes are loaded. The presentation theme loads the inner, outer and color Namkeen theme files and passes the `<options>` on to these files.

The Inner Theme

You can load the inner theme directly by
`\useinnertheme{Namkeen}`
and it has no options.



User Interface

Loading the Theme and Theme Options

The Outer Theme

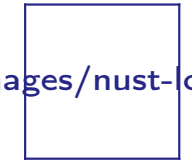
You can load the outer theme directly by

`\useoutertheme[<options>]{Namkeen}`

Currently, the theme options are

- `hidetitle`: Hide the (short) title in the sidebar
- `hideauthor`: hide the (short) author in the sidebar
- `hideinstitute`: hide the (short) institute in the bottom of the sidebar
- `shownavsym`: show the navigation symbols
- `left` or `right`: position of the sidebar (default is right)
- `width=<length>`: width of the sidebar (default is 2 cm).
- `hideothersubsections`: hide all subsections but the subsections in the current section
- `hideallsubsections`: hide all subsections

The last four options are inherited from the outer sidebar theme.



User Interface

Loading the Theme and Theme Options

The Color Theme

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`\usecolortheme{Namkeen}`

and it has no options.



User Interface

Loading the Theme and Theme Options

The Color Theme

You can load the color theme directly by
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and it has no options.

The Color Element `Namkeen`

The color theme defines a new beamer color element named `Namkeen` whose foreground and background colors are

- fg: light blue ({RGB}{194,193,204})
- bg:

You can use these colors in the standard beamer way by using the command `\usebeamercolor[<fg or bg>]{Namkeen}`. See the beamer manual for instructions.



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You can use these colors in the standard beamer way by using the command `\usebeamercolor[<fg or bg>]{Namkeen}`. See the beamer manual for instructions.
Note that this version of the theme follows official NUST Colors. However, you can easily change it (including the logo) by following the steps in `beamercolorthemeNamkeen.sty`.



Compiling Your Presentation With Namkeen Theme

Unlike most other beamer themes, this theme must be compiled at least three times to make everything look right. For most other themes, you do not have to compile your presentation more than two times. For the Namkeen theme, the third compilation is necessary to determine the position of the circle with the current frame number and the length of the "progress bar".



User Interface

Modifying the Theme

- The default configuration of fonts, colors, and layout complies with the NUST Branding - Logo and Text and is the unofficially official version of the theme.



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- However, you can modify specific elements of the theme through the templates provided by the beamer class. Please refer to the beamer user manual for instructions.



User Interface

Modifying the Theme

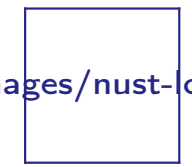
- The default configuration of fonts, colors, and layout complies with the NUST Branding - Logo and Text and is the unofficially official version of the theme.
- However, you can modify specific elements of the theme through the templates provided by the beamer class. Please refer to the beamer user manual for instructions.
- For example, on this slide the following commands have been used
  - Change the bar and sidebar colors:
    ```
    \setbeamercolor{nust}{fg=gray!50,bg=gray}
    \setbeamercolor{sidebar}{bg=red!20}
    ```
  - Change the color of the structural elements:
    ```
    \setbeamercolor{structure}{fg=red}
    ```
  - Change the frame title text color:
    ```
    \setbeamercolor{frametitle}{use=structure,
    fg=structure.fg}
    ```
  - Change the background color of the text
    ```
    \setbeamercolor{normal text}{bg=gray!20}
    ```



The Background Image

- In this documentation, the title page frame and the last frame have the waves as the background image. The waves background image can be added to any single frame by wrapping a frame in the following way

```
{\wavesbg
\begin{frame}[<options>]{Frame Title}{Frame
Subtitle}
...
\end{frame}}
```



The Background Image

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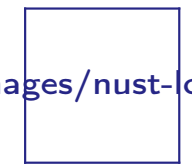
```
{\wavesbg
\begin{frame}[<options>]{Frame Title}{Frame Subtitle}
...
\end{frame}}
```

- Ideally, I would like to create a new frame option called `wavesbg` which can enable the waves background. However, I have not been able to figure out how such an option can be added.



Widescreen Support

Newer projectors and almost any modern TV support a widescreen format such as 16:10 or 16:9. Beamer (>= v. 3.10) supports various aspect ratios of the slides. According to section 8.3 on page 77 of the Beamer user guide v. 3.10, you can write
`\documentclass[aspectratio=1610]{beamer}`
to get slides with an aspect ratio of 16:10. You can also use 169, 149, 54, 43 (default), and 32 to get other aspect ratios.



Feedback

Contact Information

In case you have any comments, suggestions or have found a bug, please do not hesitate to contact me. You can find my contact details below.

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Thank you for using this theme!